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We are dedicated to helping GTU Computer Engineering students achieve academic excellence by providing university-focused Previous Year Question Papers with step-by-step premium solutions.

About This Platform

GTUStudentHub is specifically built for GTU Computer Engineering students to make exam preparation smart, easy and effective. We provide high-quality, accurate and exam-oriented solutions prepared by experienced seniors and subject experts.

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- 100% Accurate PYQ Papers
- Step-by-Step Solutions
- Exam Focused Content
- Secure & Reliable
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Our Focus

GTU Computer Engineering | Semester 6

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Semester 6 Focused

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Data Visualizations (3160717)

Summer – 2025 – Solution

Question : 01

Q.1 (a) What is AJAX in context of data reading? Explain use of it with an example.

AJAX stands for **Asynchronous JavaScript and XML**. It is used to read data from the server without refreshing the whole webpage. AJAX helps webpages load data faster and improves user interaction.

Uses of AJAX

- Read data from server
- Update webpage dynamically
- Send and receive data in background
- Improve webpage speed

Example

```
<script>
function loadData() {
  var xhttp = new XMLHttpRequest();

  xhttp.onreadystatechange = function() {
    if (this.readyState == 4 && this.status == 200) {
      document.getElementById("demo").innerHTML = this.responseText;
    }
  };

  xhttp.open("GET", "data.txt", true);
  xhttp.send();
}
</script>

<button onclick="loadData()">Load Data</button>

<div id="demo"></div>
```

In this example, data is loaded from data.txt without reloading the webpage.

Q.1 (b) Write the steps of styling your data using XSLT taking example data of your choice.

XSLT stands for **Extensible Stylesheet Language Transformations**. It is used to convert XML data into formatted HTML pages.

Steps of Styling Data using XSLT

1. Create an XML file containing data.
2. Create an XSLT stylesheet file.
3. Link XML file with XSLT file.
4. Apply HTML tags and styles in XSLT.
5. Open XML file in browser to see formatted output.

Example XML File

```
<students>
  <student>
    <name>Rahul</name>
    <marks>85</marks>
  </student>
</students>
```

Example XSLT File

```
<xsl:template match="/">
<html>
<body>

<h2>Student Data</h2>

<table border="1">
<tr>
<th>Name</th>
<th>Marks</th>
</tr>

<xsl:for-each select="students/student">
<tr>
<td><xsl:value-of select="name"/></td>
<td><xsl:value-of select="marks"/></td>
</tr>
</xsl:for-each>

</table>

</body>
</html>
</xsl:template>
```

This XSLT file converts XML data into a table format.

Q.1 (c) What are Charting Primitives? Explain any three chart types with example of each.

Charting primitives are the basic graphical elements used to create charts and graphs. Examples include:

- Lines
- Bars
- Circles
- Axes
- Labels
- Colors

These elements help in representing data visually.

1. Bar Chart

A bar chart uses rectangular bars to compare data values.

Example

Product	Sales
A	50
B	80

Uses

- Comparing categories
- Showing quantity differences

2. Pie Chart

A pie chart displays data in a circular form divided into sectors.

Example

Company	Market Share
A	40%
B	60%

Uses

- Showing percentage distribution
- Representing proportions

3. Line Chart

A line chart connects data points using lines.

Example

Monthly temperature changes:

- Jan = 20°C
- Feb = 24°C
- Mar = 28°C

Uses

- Showing trends over time
- Analyzing continuous data